agentAI: A Research-Data Platform for Physical and Embodied Agents

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A researcher states a goal; a planning model proposes the project and its agents, and the researcher corrects it. The agents then work on data they harvest, take from a robot as it drives, or receive by upload, and train on it round after round unwatched, so the loop's own report is the only account of what it did. This sheet shows that loop and the review that checks its report.

Weed Detection



The corpus exists to train the weed detector the laser robot needs. Two agents work it on their own: one harvests and audits public imagery, the other trains. Over 150,000 labelled images.

Rover



A four-wheel skid-steer narrow enough for a crop row, carrying its own battery and compute. Each drive is registered as a dataset while it runs: camera, GPS and two inertial units, over one API key.

Laser Weed Control Robot



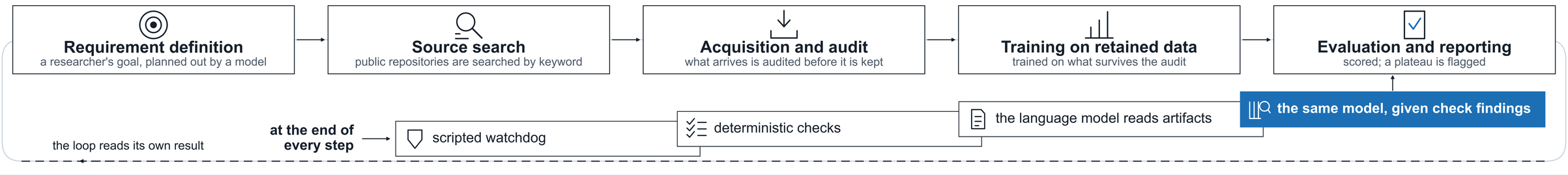
The end use the collection is for. A YOLO detector localises each weed and two galvanometer lasers aim at it and fire, destroying it in place—no chemical, and nothing touches the crop beside it.

Humanoid and Quadruped



Legs cross ground that slows wheels, and in the field this one outpaced the laser robot. A wide stance and a level gait hold a collector's camera steady.

Four projects, one platform. Each was opened by the same procedure, and its data is registered the same way whether it was harvested, driven in by a robot, or uploaded.



Agent Workflow Diagram

Figure 1. The harvest-and-train loop, as it runs unattended. A plateau is flagged today; stopping on it is a reviewer's judgement by design.

ONE PLATFORM SUPPORTS ANY ROBOT OR DATASET

Adding a robot is a configuration change

A robot, drone or sensor rig is a project: its own dataset registry, its own agents, and an owner who controls both. Agents harvest, filter, label and train against that registry.

The weed detector runs on a robot's live view

While a robot drives, its project page shows the newest camera frame every second and a half. One switch runs the weed detector over that frame, so the operator sees its detections during the drive.

Ask any dataset a question in plain words

Every dataset page has a chat box. A planner reads the question and the data's real columns, then picks a built-in analysis or writes new code. That code runs in a sandbox of whitelisted libraries, and the answer returns with its chart and code.

Scan to open the live platform

lab-b660m-c.tailfa6424.ts.net

Sign in with a Google account to open it.

OPEN-WEIGHT MODELS REVIEW IN TIERS

A flag is not a citation

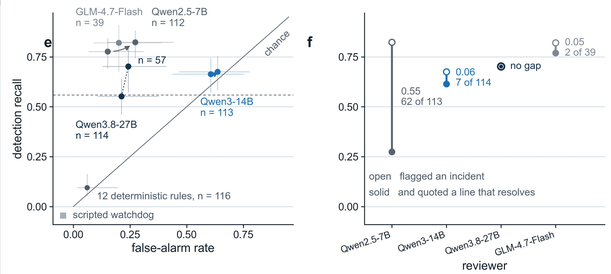


Figure 2. (e) Recall and (f) grounded recall on 149 frozen cases, 116 of them failures; each arrow runs from a model on raw files to the same model given the checks' findings. A dotted arrow's ends were scored on different cases, 57 against 114.

Qwen2.5-7B flags 0.82 of the failures but quotes a line that resolves only 0.27. The larger models cite nearly everything they flag.

Review runs even when nothing looks wrong

Review follows every step, because failures here announced themselves as successes. Four levels, by cost: the running watchdog, twelve fixed checks, a model reading raw files, then that model given the checks' findings.

WHEN A MODEL'S JUDGEMENT COUNTS

What the lab model says stays a draft

On the cluster	
plan the next step	qwen3.8 27B
review in depth	glm-4.7-flash
hard one-off reasoning	deepseek-v3 671B
decide what to collect	gemma 4
judge dataset quality	qwen2.5 7B
label assistance	minicpm-v
On the lab box	
turn intent into a project	qwen2.5 3B
summarise a dataset	qwen2.5 3B
The tiers above them	
the hardest open problems	Kimi K3, largest open
check and correct a campaign	Claude Opus 5, via Claude Code
plan the next experiments	Claude Fable 5.1, own key

Figure 3. Each job that needs a model and the model it was given, from the router's table; a filled square marks judgement. The dashed band is designed above it.

Two jobs run in the lab because a person is waiting; their answers are marked, so none can count as a decision.

What a model may do without asking

Each action is risk-graded. A reviewer may change a setting, and may submit a job once its verdict cites evidence. Costly steps need approval, and only a person may delete. The first approval went to a model's hypothesis, later disproved.

WHAT THE COLLECTION PRODUCED

Two agents collect and train side by side

A collector searches Kaggle, Hugging Face and Roboflow on its own, and a trainer takes each new dataset between rounds. The detection pool holds 179,989 images across 63 datasets.

Every image is checked on the way in

Mirrored copies are caught by image content, 65,007 in one pass. No test-set image can reach training. Each source is registered with its licence, and where a dataset had no boxes, a model draws them.

A quality gate decides what the trainer sees

A dataset joins the training pool only if its images resemble trusted weed data in DINOv2 features, which keeps off-topic sets out; fifty-one were held back. A vision model asked about single boxes said yes to every one.

Dataset filters miss single bad boxes

A labelling mistake can be one bad box in an otherwise sound picture, invisible to a whole-dataset filter. Per-box checkers are built; whether gating training on their scores helps is the next experiment.